MIL-C-46933A(MU)
22 April 1966
SUPERSEDING
MIL-C-46933(ORD)
20 September 1962
AMENDMENT 3
16 November 1964

MILITARY SPECIFICATION
CARTRIDGE, 7.62MM, NATO, BLANK, M82

1. SCOPE

1.1 This specification covers Cartridge, 7.62mm, NATO, Blank, M82 for use in 7.62mm, weapons for saluting and training.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-636	- Visual Inspection Standards for Small Arms Ammunition Through Caliber .50
MIL-STD-644	- Visual Inspection Standards and Inspection Procedures for Inspection of Packaging, Packing and Marking of Small Arms Ammunition

DRAWINGS

Munitions Command

C8597283	- Cartridge, 7.62mm, NATO, Blank, M82
F8594726	- Packing and Marking; Cartridges, 7.62mm, NATO; 5-rd Clips; Cartons; Bandoleers; Box, Ammunitions, M2A1; Box, Wirebound

FSC 1305

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DRAWINGS (Cont'd)

Munitions Command

F10521861	- Packing and Marking; Cartridges, 7.62mm, NATO, Links; Cartons; Bandoleers; Box, Ammunition, M19A1; Box, Wirebound
F10523070	- Packing and Marking, Cartridge, 7.62mm, NATO, Blank, M82; Cartons; Box, Ammunition, M2A1, Box, Wirebound
IEL8597283	- List of Inspection Equipment for Cartridge, 7.62mm, NATO, Blank, M82

PUBLICATIONS

Munitions Command

AMCR 715-505, Vol. 3	- Ammunition Ballistic Acceptance Test Methods - Vol. 3 - Test Procedures for 7.62MM Cartridges

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- The cartridge shall comply with the requirements specified on Drawing C8597283, referenced specification, and the following:

3.2 First article testing.- Requirements for submission of first article samples by the contractor shall be as specified in the contract.

3.3 Residual stress.- The cartridge case shall not split when subjected to a one percent mercurous nitrate solution for 15 minutes.

3.4 Screen perforation.- The cartridge, when fired, shall not cause perforations of 0.1 inch or greater in diameter in a paper screen placed 15 feet from the muzzle of the gun.

3.5 Muzzle flash.- The flash produced at the muzzle of the weapon shall be visible 100 yards from the weapon under conditions of darkness.

3.6 Fouling.- The fouling produced by the firing of 500 cartridges shall not cause a change in the cyclic rate by more than 15 percent from the first to the last burst.

3.7 Function and casualty.- The cartridge shall function without casualty.

3.8 Cyclic rate.- The cartridges shall operate the M60 Machine Gun at a cyclic rate of not less than 450 cartridges per minute and the M14 Rifle at a cyclic rate of not less than 600 cartridges per minute.

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3.9 Workmanship.- The requirements for workmanship are as specified on the applicable drawings, referenced specification, and the following:

3.9.1 Metal defects.- The cartridge shall be free of folds, wrinkles, deep draw scratches, scaly metal, dents and other defects.

3.9.2 Foreign matter.- The cartridge shall be free of corrosion, stains, discolorations, dirt, oil and smears of lacquer.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.2 First article sample.

4.2.1 Initial production sample.- At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 3000 cartridges. The sample shall be manufactured using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials, including packaging and packing, shall be the same as used for regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and test.- After inspection and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a government laboratory or such other facility specified in the contract.

4.2.1.2 Initial production sample failure.- Failure of the sample to comply with requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions.

4.3.1 Lot.

4.3.1.1 Submission of product.- The product shall be submitted in accordance with MIL-STD-105.

4.3.1.2 Lot identification.- Each lot of ammunition shall be identified as to type, caliber and model as well as with a lot number and the supplier's identification as assigned by the procuring activity. Each lot shall be further identified by a Federal Stock Number assigned by the procuring activity.

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4.3.2 Examination.- Examination for major and minor defects shall be performed on a class basis in accordance with the classification of defects, Table I, using applicable sampling plans and acceptance criteria of Standard MIL-STD-105. The Acceptance Quality Level (AQL) for the major class shall be 0.65 percent and the AQL for the minor class shall be 2.50 percent.

4.3.2.1 Classification of defects.- The classification of defects shall be as specified in Table I.

TABLE I

No.	Defect and method of inspection	Major	Minor
	Visual 1/		
1	Discolored, dirty, oily, smeared		X
2	Corroded or stained, if etched		X
4	Round head	X	
5	Dent		X
6	Split 2/	X	
7	Perforation	X	
8	Draw scratch		X
10	Beveled underside of head	X	
12	Scaly metal		X
13	No chamfer on head (rim)		X
14	Fold		X
18	Illegible or missing head stamp		X
19	Defective head		X
32	No primer	X	
33	Cocked primer	X	
34	Inverted primer	X	
35	Loose primer	X	
36	Nicked or dented primer		X
37	No waterproofing material (primer pocket joint)		X
38	Defective primer crimp		X
39	Missing or loose wad	X	
40	Wad and joint between neck and wad not completely covered by lacquer		X
41	Opening between wad and case wall	X	
	Gaging		
42	Cartridge profile failure (requiring more than 20 pounds dead weight to insert in profile and alignment gage)	X	
43	Diameter of mouth, min.	X	

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ble 1, Gaging (Cont'd)

No.	Defect and method of inspection	Major	Minor
44	Diameter of mouth, max.		X
45	Diameter of extractor groove, max.	X	
46	Diameter of extractor groove, min.		X
47	Diameter of head, max.	X	
48	Diameter of head, min.		X
49	Thickness of head		X
50	Length of shoulder datum, min.		X
51	Depth of primer, min.	X	
52	Depth of primer, max.		X

1/ Refer to 7.62mm section of Standard MIL-STD-636 for visual standards for defects 1 through 38. In the event of conflict between table I of this specification and MIL-STD-636 as to defect classification, the classification specified in Table I shall apply.

2/ A split shall be classed as a major defect regardless of location or occurrence of loss of propellant.

4.3.3 Tests.- The tests listed in Table II shall be conducted in accordance with the methods and procedures specified in 4.4.

4.3.3.1 Test samples.- The quantities for the various tests shall be as specified in Table II. Only cartridges having met the visual and dimensional requirements shall be used in the ballistic tests, and shall have been selected in such a manner that the sample is representative of the entire lot. The cartridges selected shall be thoroughly mixed before being divided into samples for the various tests.

TABLE II

Test	Number of Cartridges	Requirement Paragraph
Residual stress (mercurous nitrate 1/	50	3.3
Screen Perforation 2/	200	3.4
Flash 3/	20	3.5
Fouling 3/	500	3.6
Function and Casualty 4/		3.7
Gun, Machine, 7.62mm, M60	400	
Rifle, 7.62mm, M14	240	
Cyclic Rate 5/		3.8

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1/ Failure of two or more cartridges to comply with the applicable requirement shall be cause for rejection of the lot. If one cartridge fails in the first test, a second sample consisting of double the number of cartridges in the first sample shall be tested. If any failing cartridges are found in the second sample, the lot shall be rejected.

2/ The lot shall be rejected if more than three cartridges fail to comply with the applicable requirement, subject to the selection of a second sample of double the quantity of the initial sample, provided that not more than eight cartridges fail to comply with the requirement in the initial test. The lot shall be rejected if in the combined test of the first and second sample, nine or more cartridges fail to comply with the applicable requirement.

3/ This test shall be conducted on the initial production sample only.

4/ An initial production sample or production lot shall be rejected when function and casualty defects plus all other firing defects (excluding fouling tests) exceed the acceptance number for the cumulative sample in Table III. If the number of defects found in the first test exceeds the acceptance number for the first sample but is equal to or less than the acceptance number for the cumulative sample, a second sample consisting of double the quantity specified under function and casualty test shall be fired in the service weapon specified therefor. This procedure shall apply regardless of the weapon or weapons in which the firing defects occurred in the first tests. If the total number of defects in the combined first and second sample exceeds the acceptance number for the cumulative sample, the lot shall be rejected. If, in testing a second sample, defects other than those for which the second sample is being tested should occur to the extent that they exceed the acceptance number for the cumulative sample, the lot shall be rejected.

5/ The cyclic rate shall be determined during the firing of the function and casualty test. Failure of any burst to comply with the applicable requirement shall be cause for rejection of the lot subject to the testing of a second sample consisting of double the number of cartridges in the first sample. Failure of any burst in the second sample to comply with the applicable requirement shall be cause for rejection of the lot. The second sample shall be fired in the type weapon in which the failure occurred.

4.3.3.2 Firing defects.- Firing defects and acceptance numbers shall be specified in Table III.

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TABLE III

Defects	Acceptance Numbers (First Sample)	Cumulative (First & Second Sample)
1. Primer		
a. Escape of gas through or around primer cup	12	32
b. Blown primer or primer falls out of pocket on retraction of bolt	0	1
2. Case		
Splits, neck, shoulder or mouth	12	32
Bulges, mouth or neck	12	32
Splits, body	4	10
Splits, through head	1	2
Complete rupture	0	1
Partial rupture	4	10
Detached metal 1/	0	1
3. Weapon Stoppage 2/	12	32
4. Misfire	3	8

1/ Any particle of metal which becomes separated from the case between the mouth and the juncture of the shoulder and neck shall be classified as detached metal.

2/ All stoppages attributable to the ammunition, with the exception of misfire or complete rupture observed in all tests shall be included.

4.3.4 Packaging, packing, and marking inspection.- During or immediately prior to the packaging operation, 100% examination of the cartridges shall be performed to ascertain that the cartridge type conforms to the drawing. Occurrence of another model blank cartridge or a dummy cartridge shall be classed as a major defect. Occurrence of any type other than those listed shall be classed as a critical defect. All nonconforming cartridges shall be rejected. Inspection for packaging, packing, and marking shall be in accordance with MIL-STD-644 as applicable to the drawing.

4.3.5 Inspection equipment.- The examination and tests shall be made using equipment listed on IEL-8597283, except as provided in 4.3.

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4.4 Test methods and procedures.

4.4.1 Residual stress (mercurous nitrate).- Tests shall be conducted in accordance with AMCR 715-505, Volume 3.

4.4.2 Screen perforation.- The test shall be conducted in accordance with Publication AMCR 715-505, Vol. 3.

4.4.3 Muzzle flash.- The test shall be conducted on a darkened indoor range. The cartridges shall be fired single shot, and observation for flash shall be made from a safe vantage point at a distance of 100 yards from the muzzle of the weapon.

4.4.4 Fouling.- The cartridges shall be fired in bursts of 50 cartridges and the weapon shall be cooled to ambient temperature after the firing of every 100 cartridges. The cyclic rate of the first and last burst shall be determined and recorded.

4.4.5 Function and casualty.- In these firings the weapons shall be at room temperature at the beginning of the test. Two machine guns and two rifles shall be used and the following number of cartridges shall be fired in each weapon:

Rifle	120
Machine Gun	200

The test shall be conducted in accordance with Publication AMCR 715-505 Vol. 3, with the following exceptions:

a. The machine gun shall be fired in bursts of 50 cartridges each.
b. In all rifle firing, magazines containing 20 cartridges each shall be fired alternately semi-automatic and full-automatic.

4.4.6 Cyclic rate.- The cyclic rate shall be determined during the function and casualty test. The cyclic rate of each burst in the machine gun and each magazine fired full automatic in the rifle shall be recorded.

4.4.7 Defect penalty.- In any ballistic test, except function and casualty, in which the occurrence of a firing defect prevents the obtaining of a reliable result for the characteristic being tested, an additional shot shall be fired. That particular test shall not be penalized, but the acceptance or initial production sample shall be penalized for such defects in accordance with Table III.

5. PREPARATION FOR DELIVERY

5.1 Packing.- Level A (Worldwide shipment).- The cartridge shall be packed in accordance with Drawing F8594726, F10521861 or F10523070.

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5.2 Marking and labeling.- Packing boxes shall be marked and labeled in accordance with the applicable drawings cited in 5.1.

6. NOTES

6.1 Ordering data.- Invitations for bids and contracts or orders will specify the following:

6.1.1 Title, number and date of this specification.

6.1.2 Type and level of packing.

6.1.3 Provision for the supply, maintenance, and disposition of mandatory ballistic test equipment for acceptance inspection purposes.

6.1.4 Provision for the submission of acceptance inspection reports containing final inspection results for each lot of ammunition presented to the Government.

6.1.5 Requirement for contractor to provide and maintain an inspection system in accordance with MIL-I-45208, Inspection System Requirements.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1305-A434

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT

$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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